

Imagine its 3AM

- **Ramesh, 74, wakes up to use the bathroom.**
- He doesn't turn on the lights — doesn't want to wake his wife.
- Three steps from the bed, his foot catches the rug.
- He falls. Hard.
- His phone is charging in the other room.
- His wife is a heavy sleeper. He lies there for 4 hours.
- By the time help arrives, a simple fall has become a life-threatening emergency.

This happens 6,84,000 times a year. ¹

1 in 5 elderly fall victims die within a year of injury. ¹

Average time to receive help after a fall: 2+ hours.

Every minute of delay increases mortality by 3%. ²

Source: 1. WHO Global Report on Falls Prevention 2023; 2. Centre for Disease Control and Prevention (CDC), Older Adult Fall Prevention, 2022.

Now, imagine its 9PM

- **Lakshmi, 30, is walking home from the metro.**
- She hears footsteps behind her. She quickens her pace.
- Her phone is buried at the bottom of her bag.
- By the time she finds it, it's too late to dial.
- Three streets away, her son Gopal, 9, hasn't come home from his tuition.
- He left 90 minutes ago. The tuition is 15 minutes away.
- Priya doesn't know where he is.

88% of Indian women feel unsafe in public spaces. 3

A woman is sexually assaulted every 15 minutes in India. 4

45,000+ children go missing in India every year. 5

Source: 3. International Center for Research on Women (ICRW), Safety Audit of Public Spaces, India, 2021

4. NCRB (National Crime Records Bureau) Annual Report 2022 5. NCRB Missing Persons Report 2023



Safe Shade

Your Everything Safety Companion

Pitchdeck Presentation for
5G Innovation Hackathon

June, 2026

Patent
Pending

The world's first adaptive 5G safety wearable that protects **everyone**.

Your **grandmother**. Your **sister**. Your **children**.

And **calls for help without your phone**.

No other existing product automatically re-calibrates safety thresholds based on the context and who's wearing.



SafeShade 5G



ADVANCED

3-LAYER FALL DETECTION

Freefall + Impact + Stillness



MEDICAL ID

NFC + QR
Emergency Access

5G

5G / LTE-M
CONNECTIVITY
Standalone
cellular operation



GUARDIAN MESSAGING

Two-way communication
& safety checks



HAPTIC & AUDIO ALERTS

Reminders and
notifications

UP TO
7 DAYS
BATTERY LIFE

Deep sleep optimization
§ Estimated for SafeShade 5G



BLE NETWORK LOCATION

Find with or without
your phone



SMART HOME INTEGRATION

Matter protocol enabled §
§ Planned

PRECISION

MULTI-GNSS LOCATION

GPS + Galileo +
GLONASS + BeiDou
Auto Geo-Fencing



INSTANT

AUTO DISTRESS CALLING

Caches closest emergency
services contacts and
notifies independently



HEALTH MONITORING §

Heart Rate + SpO₂ +
Temperature

§ Planned



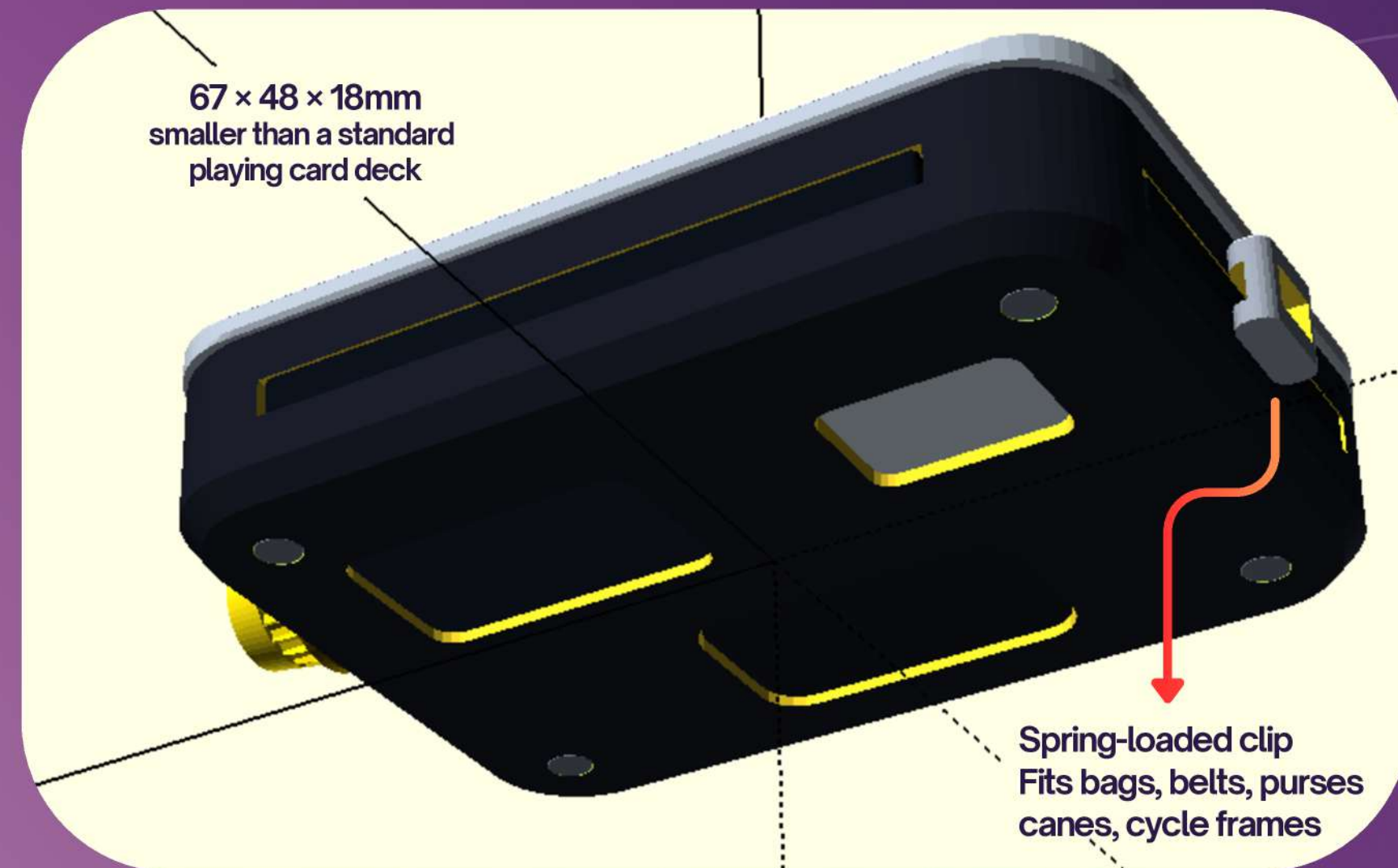
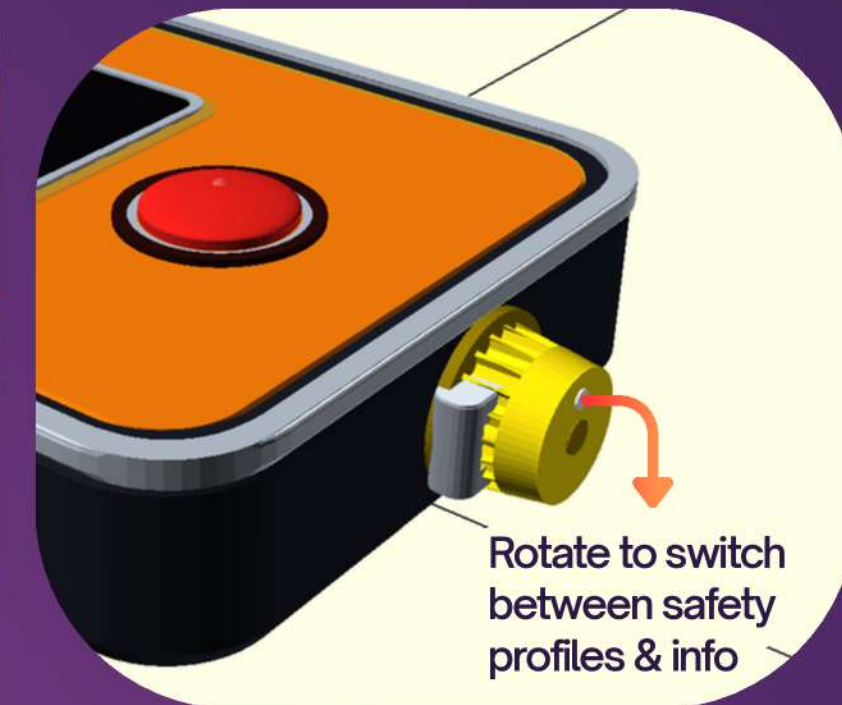
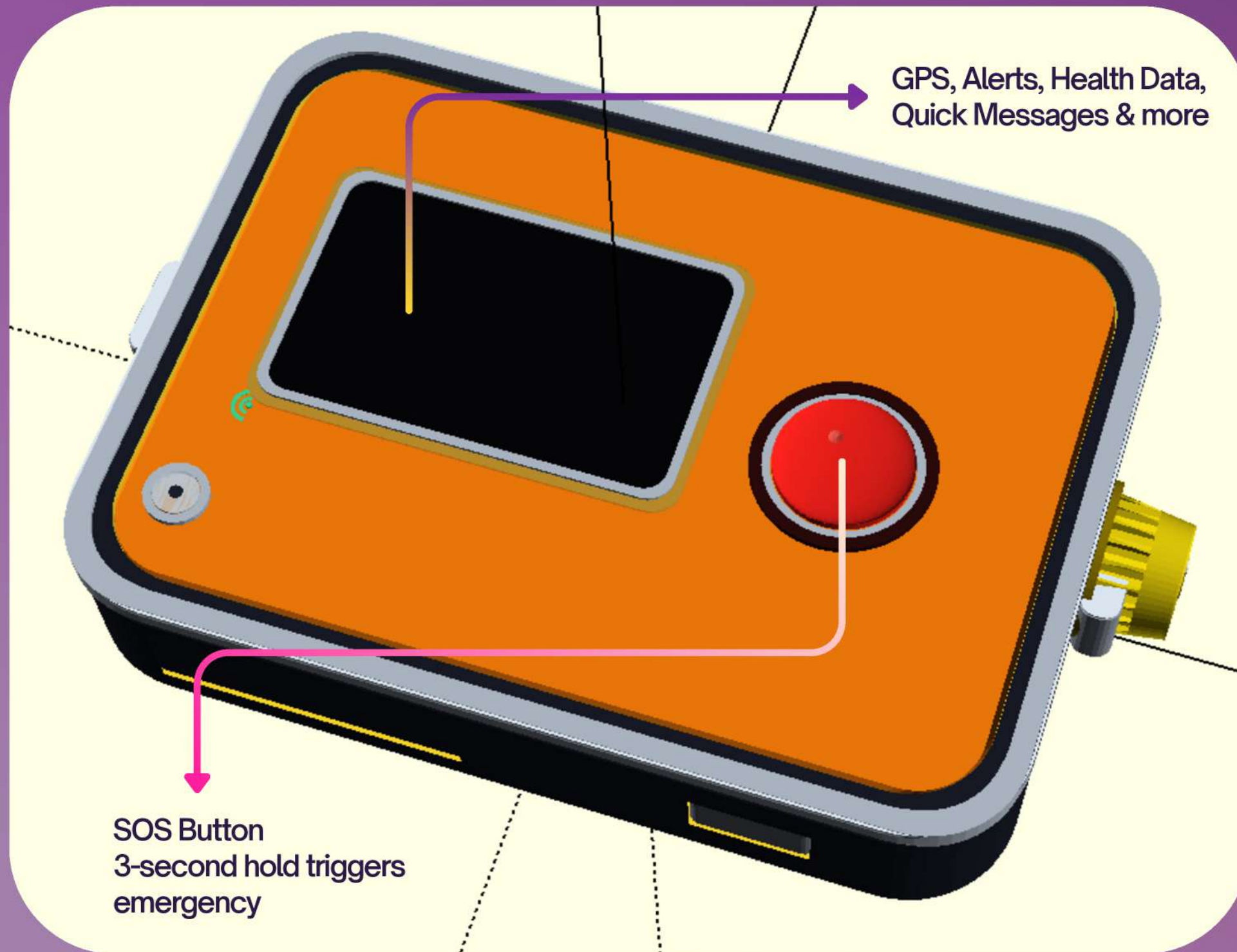
SAFESHADE SPARK

Companion Panic-Button Wearable
Standalone & Auto SOS Activation

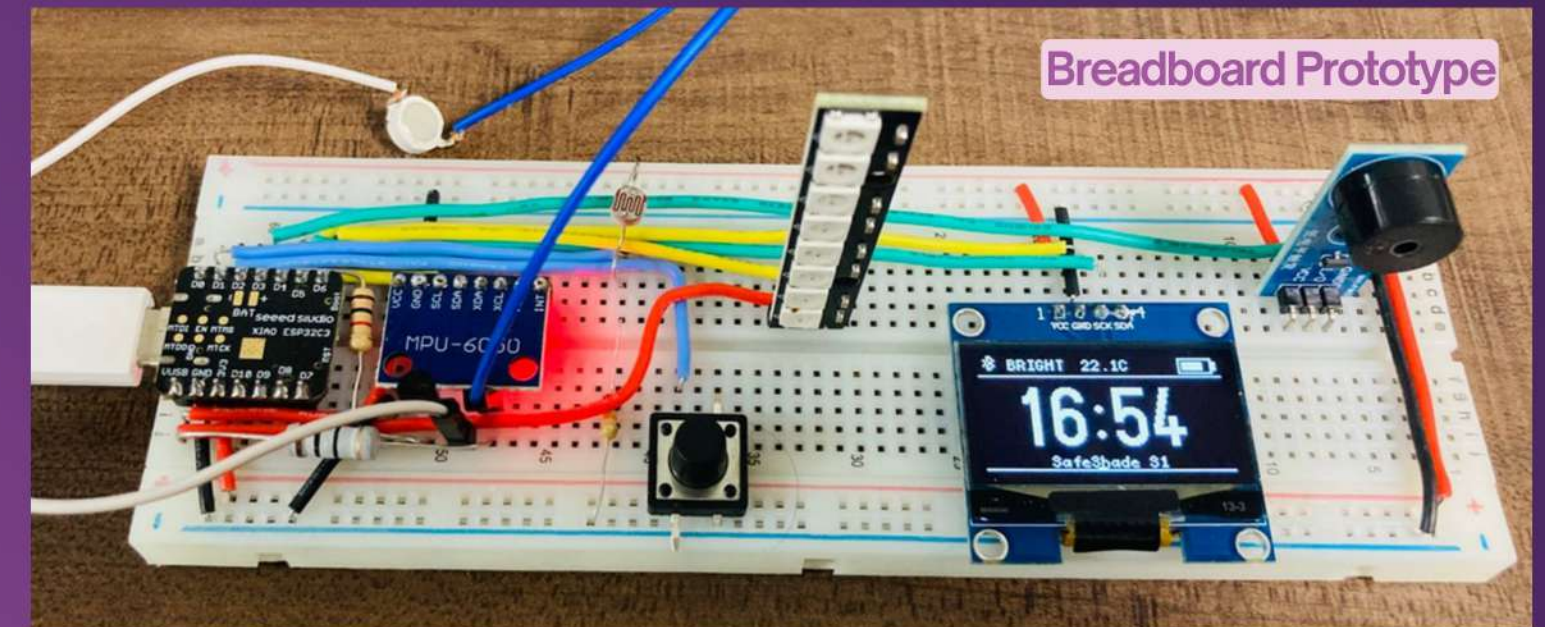


Safe Shade

5G



TRL-4 Prototype, Android App, Custom PCB & Enclosure Design



Home Screen



Weather Data



Live GPS



Messages



Quick Reply



Message Thread



LED Alerts



Emergency Active

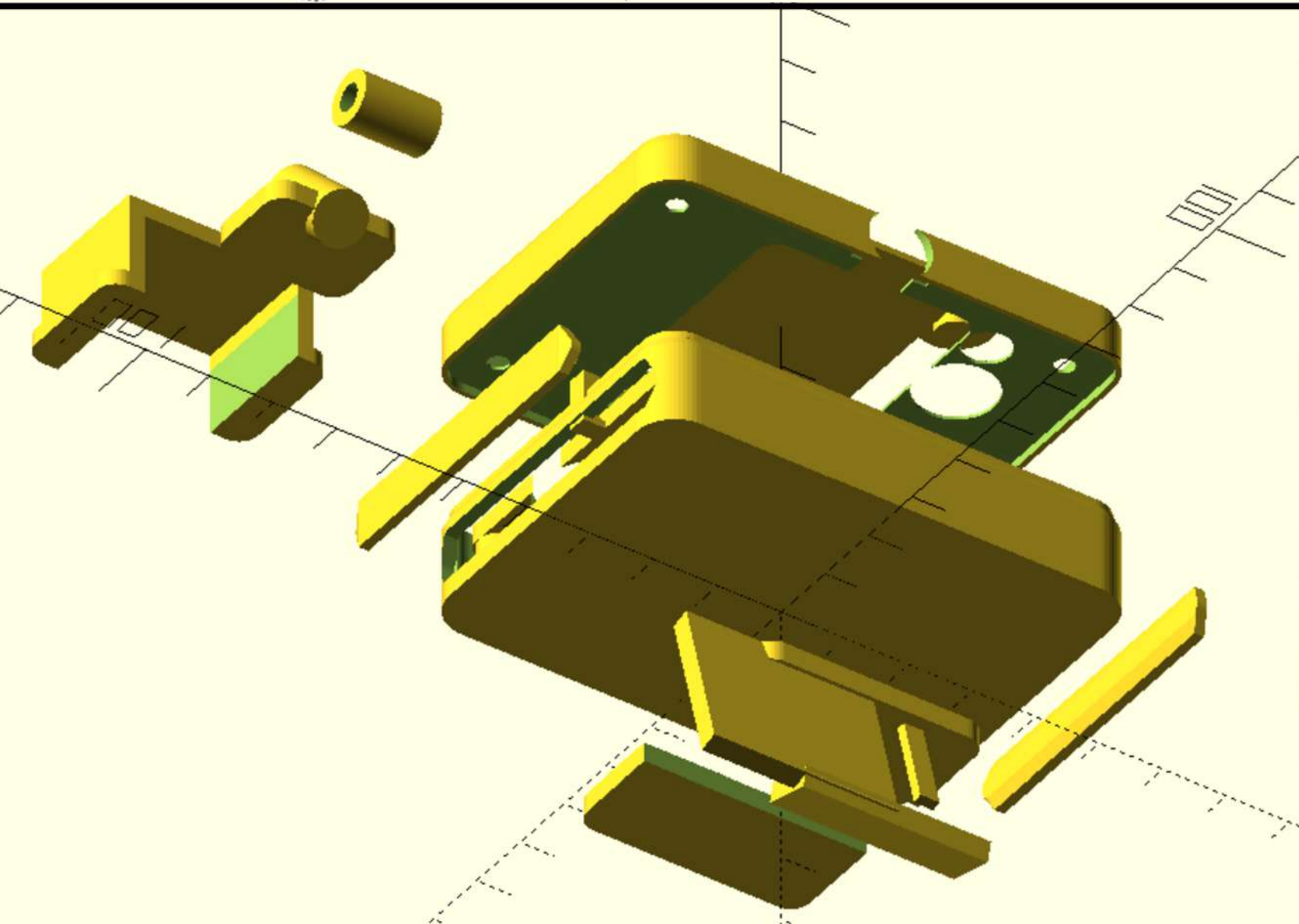
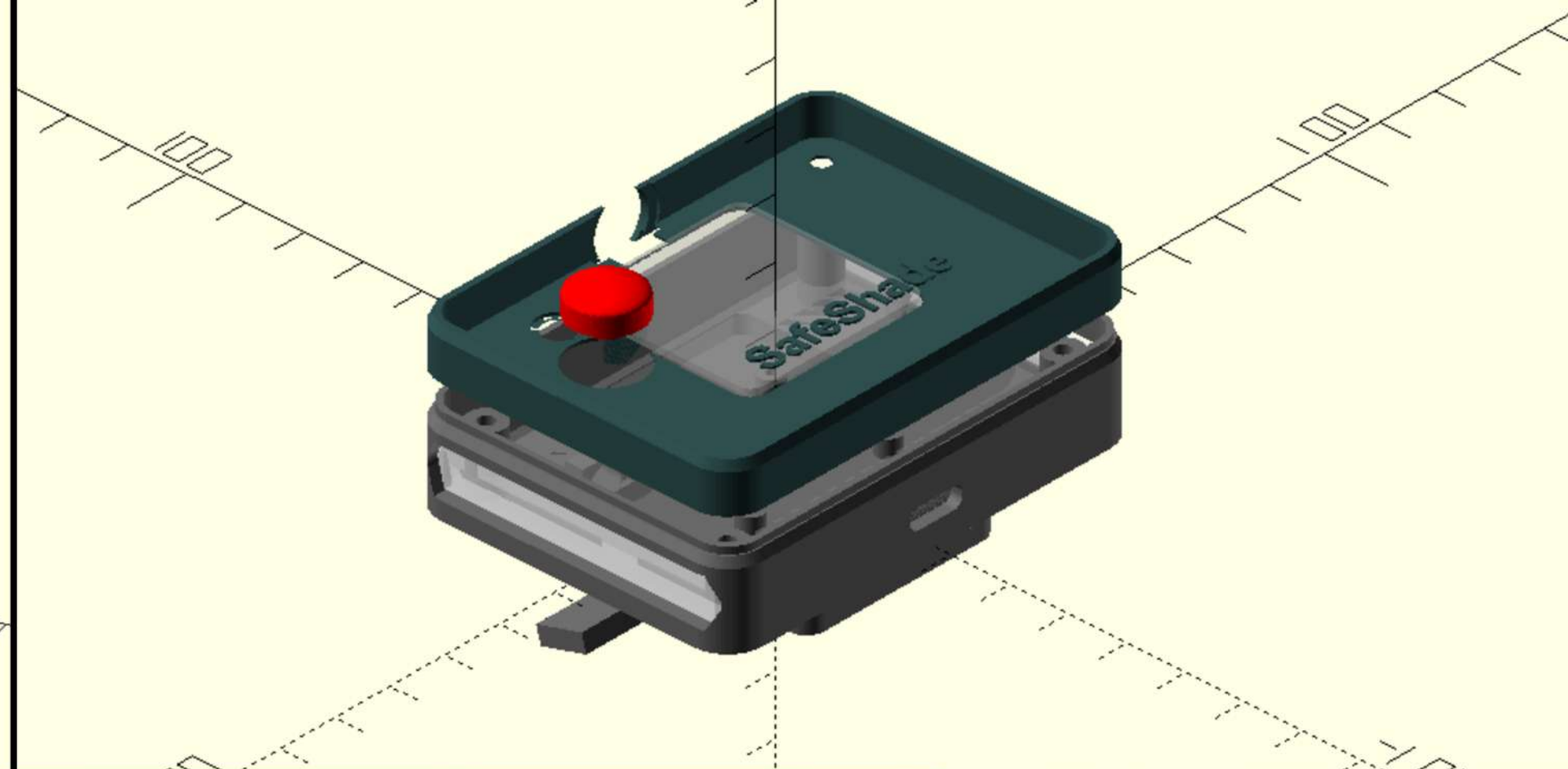
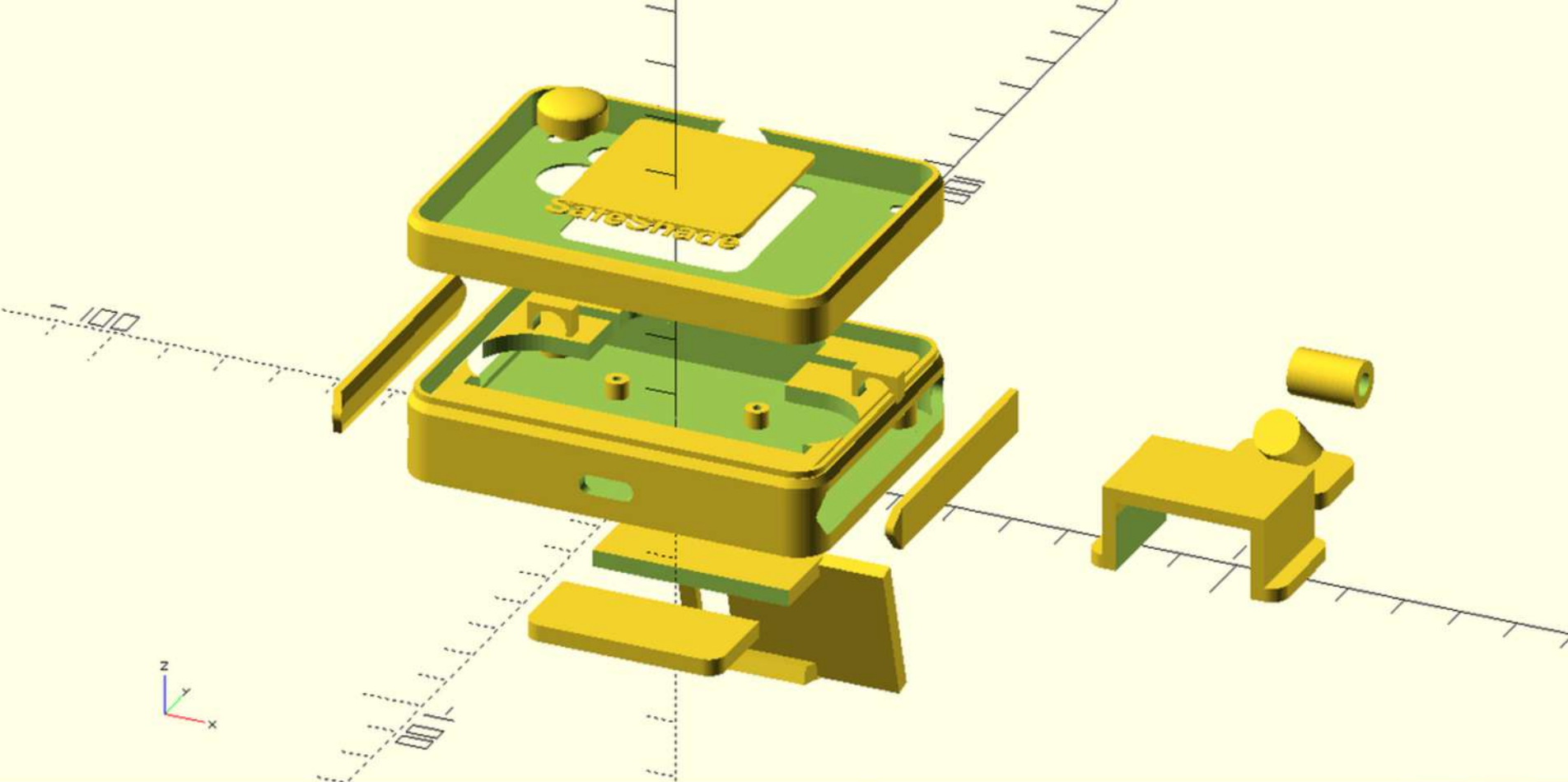


Fall Detected

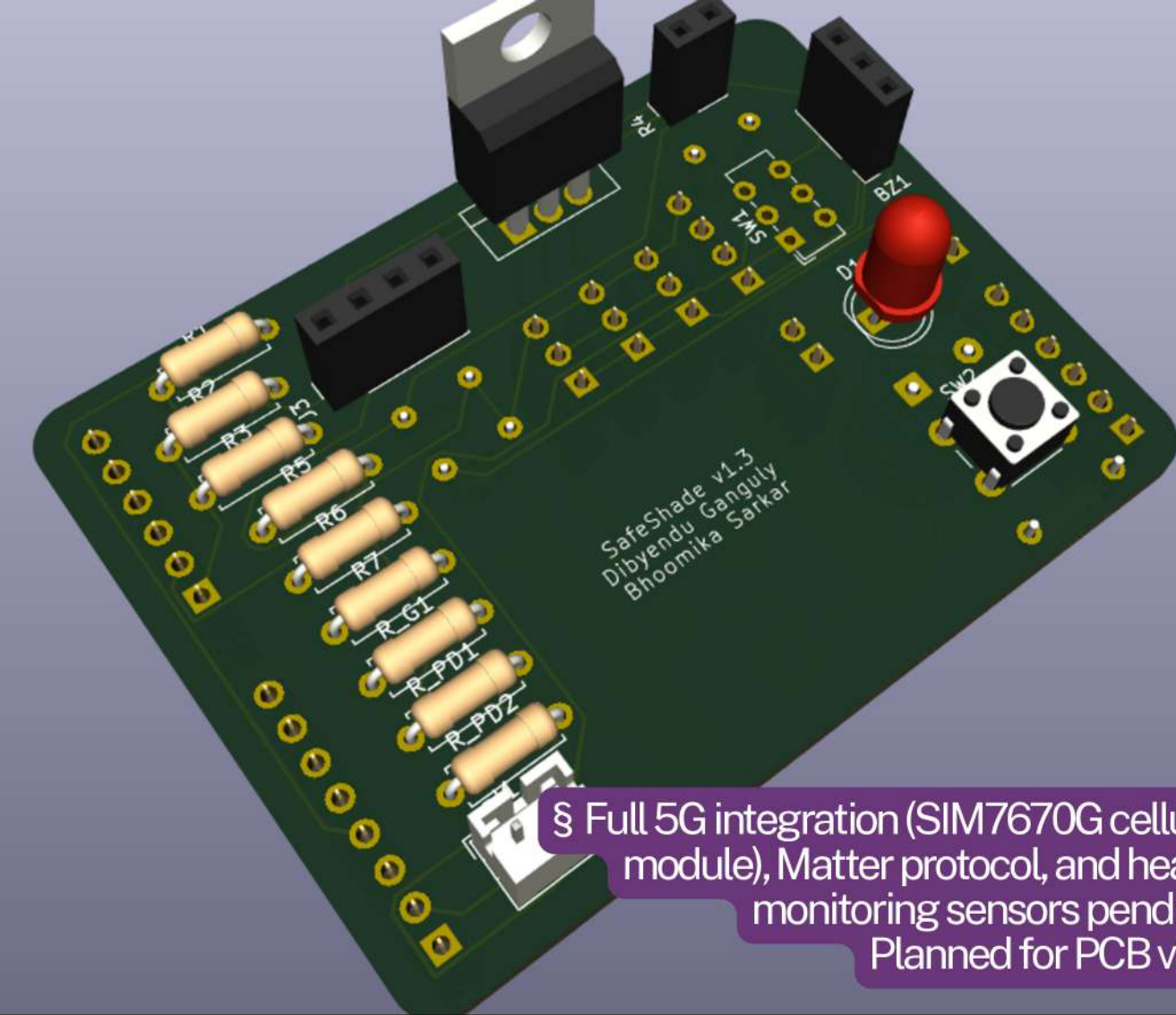


Medical ID

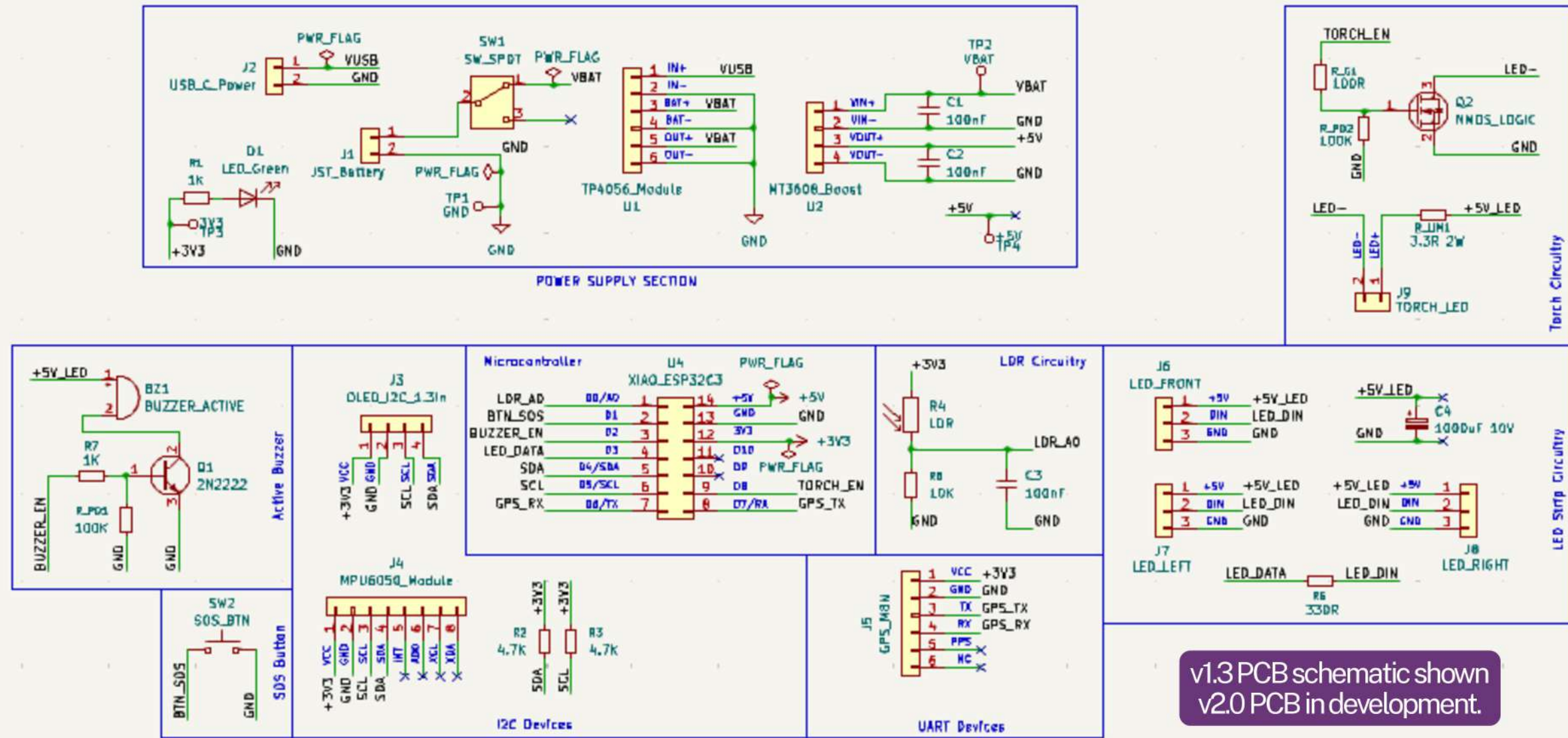
§ TRL-4: Technology validated in controlled laboratory environment. Full 5G integration & network access pending.



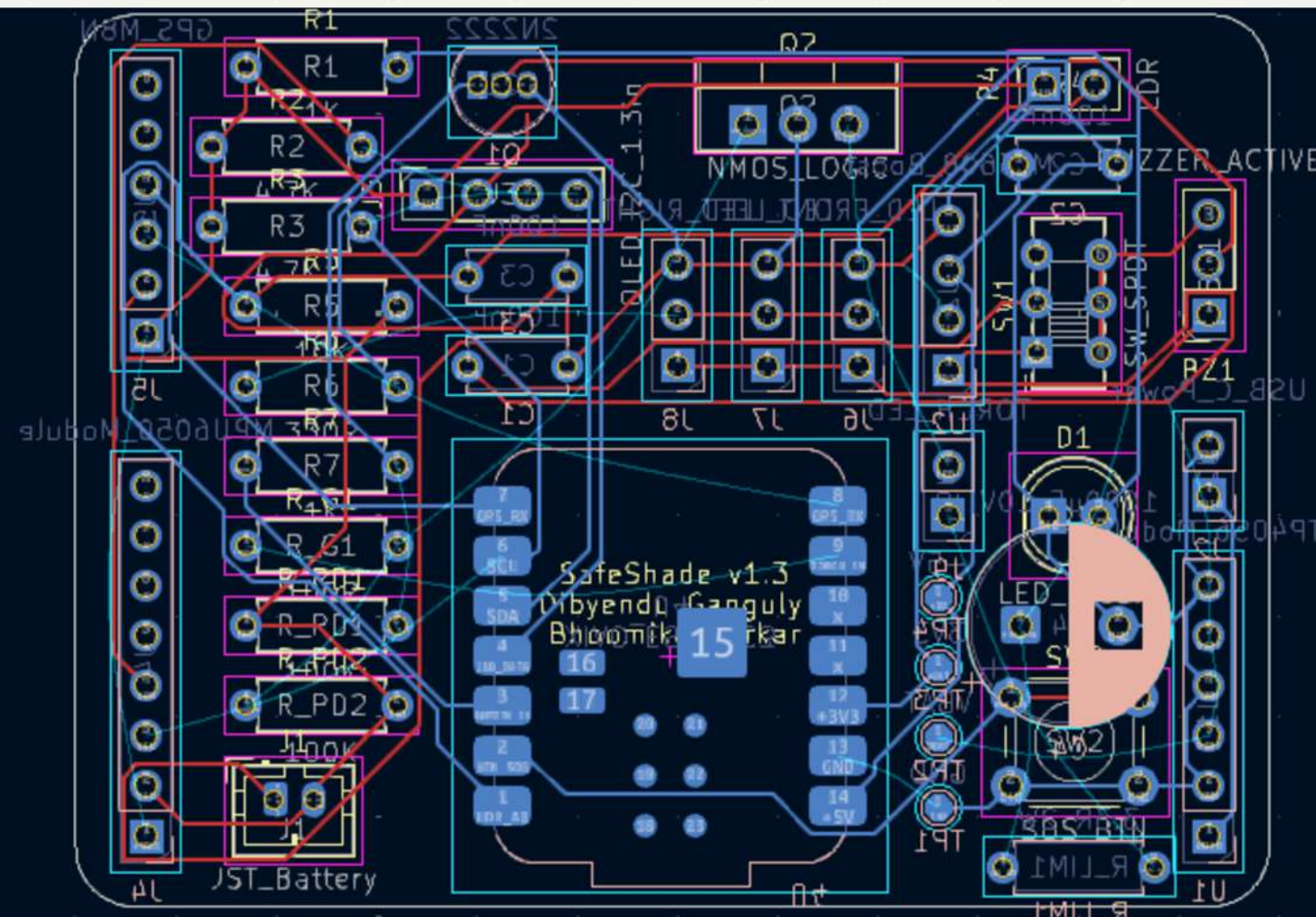
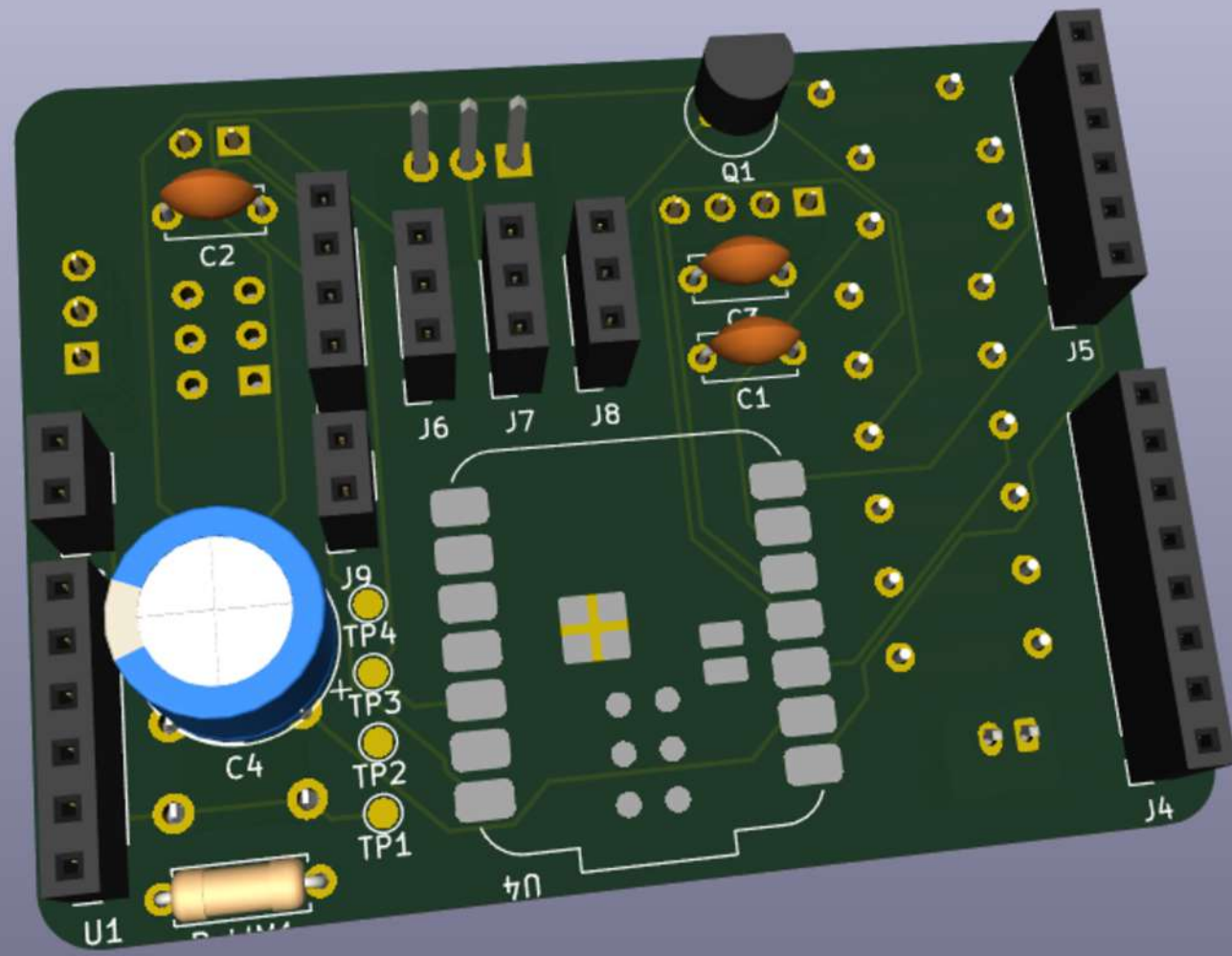
- **1. Compact Universal Form Factor** - 67 × 48 × 18 mm — Clips to backpacks, belts, canes, bike frames
Spring-loaded clip fits straps up to 50mm width
- **2. Functional Integration** - 270° LED diffuser channels for front/left/right visibility
Dedicated OLED window, acoustic grille for 85dB+ , output, Tactile button with raised profile for easy access LDR window.

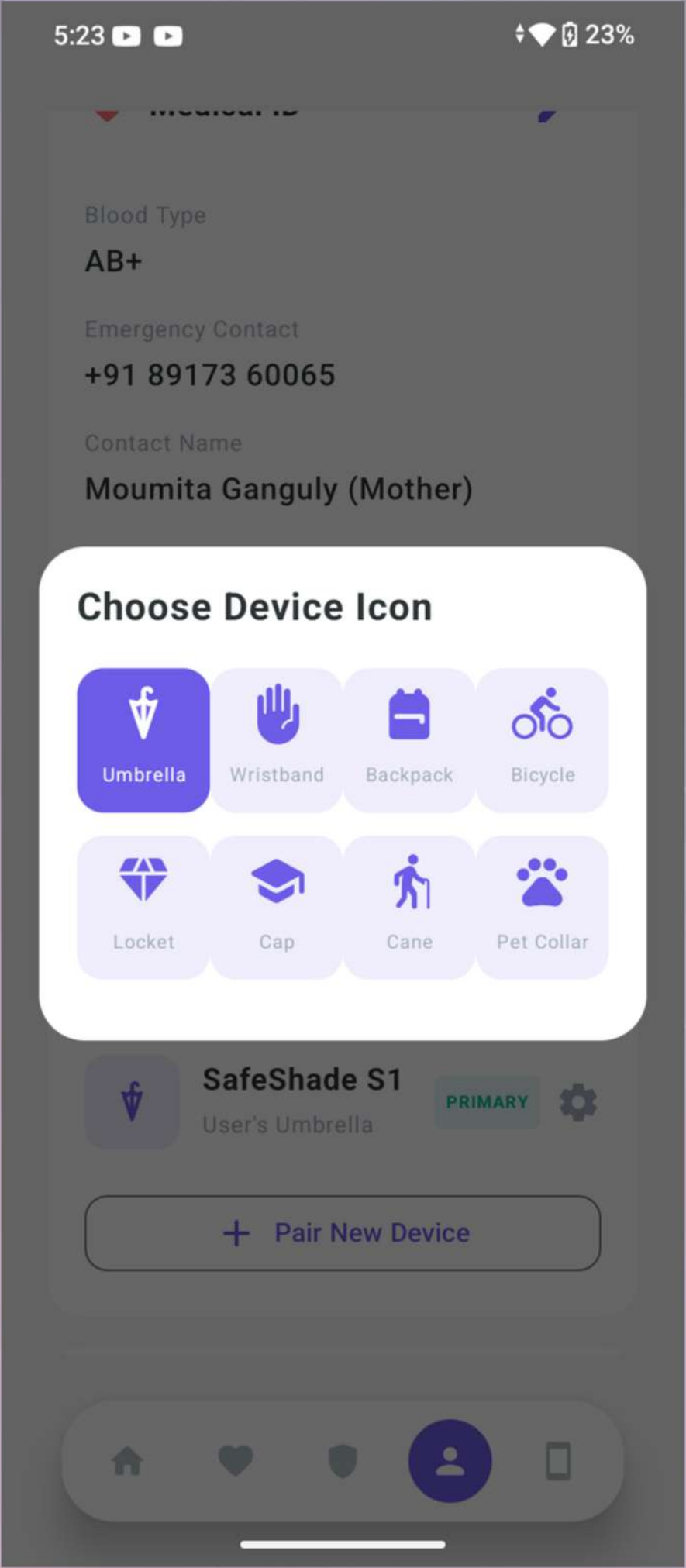
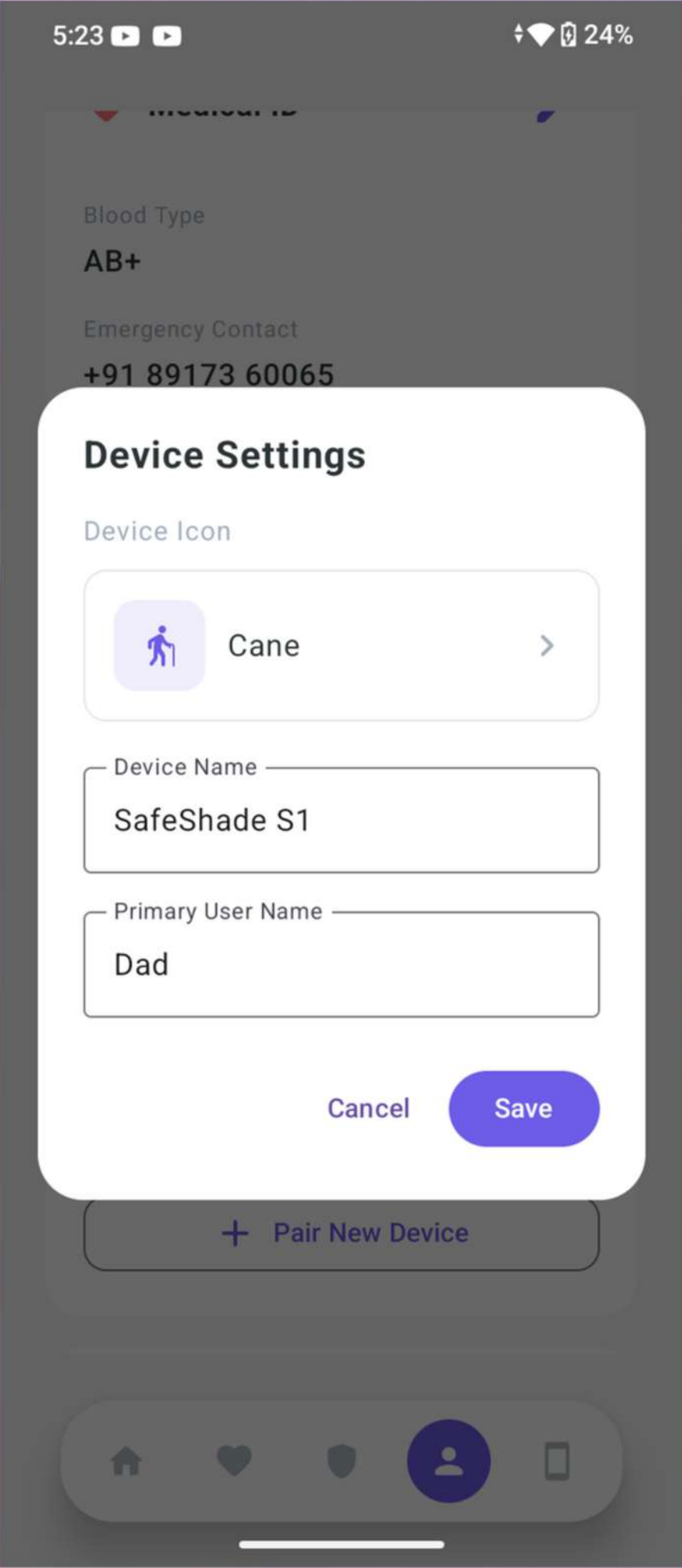
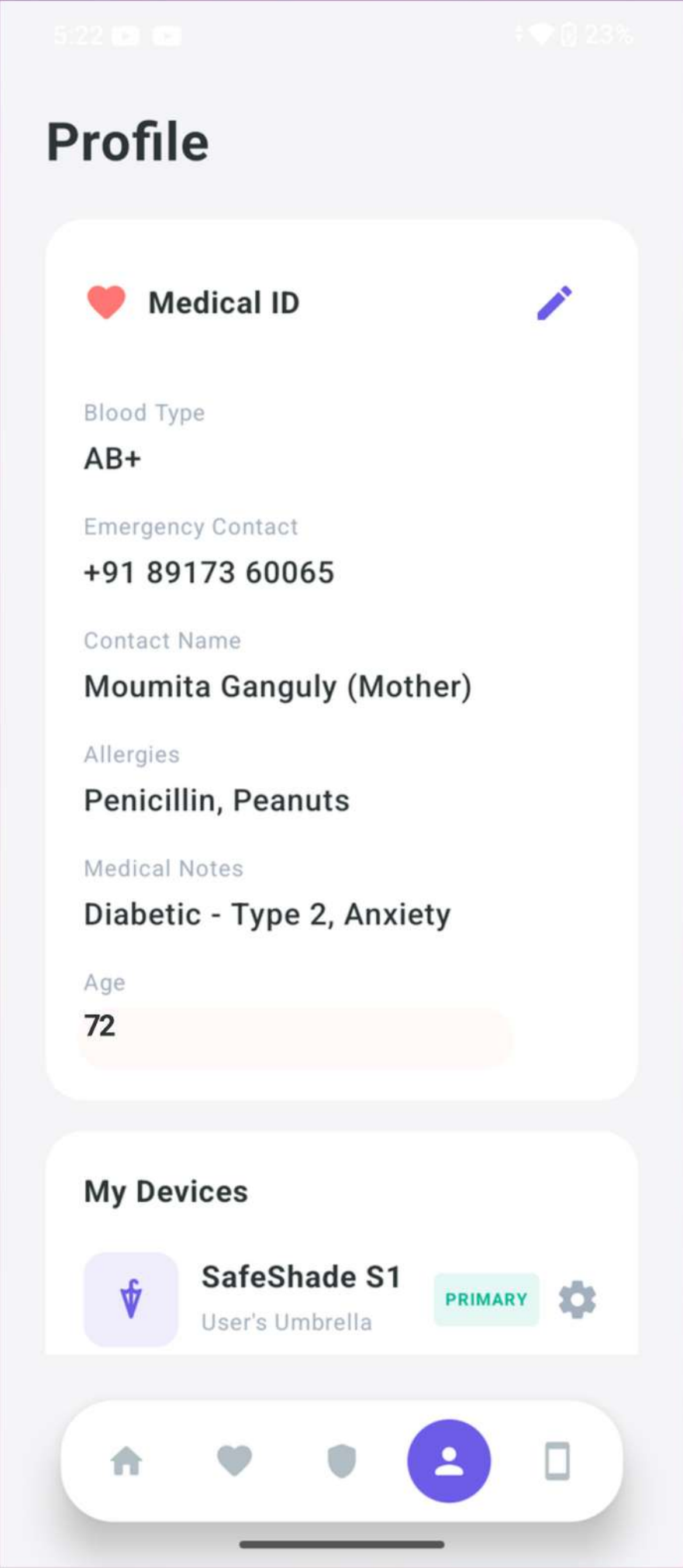
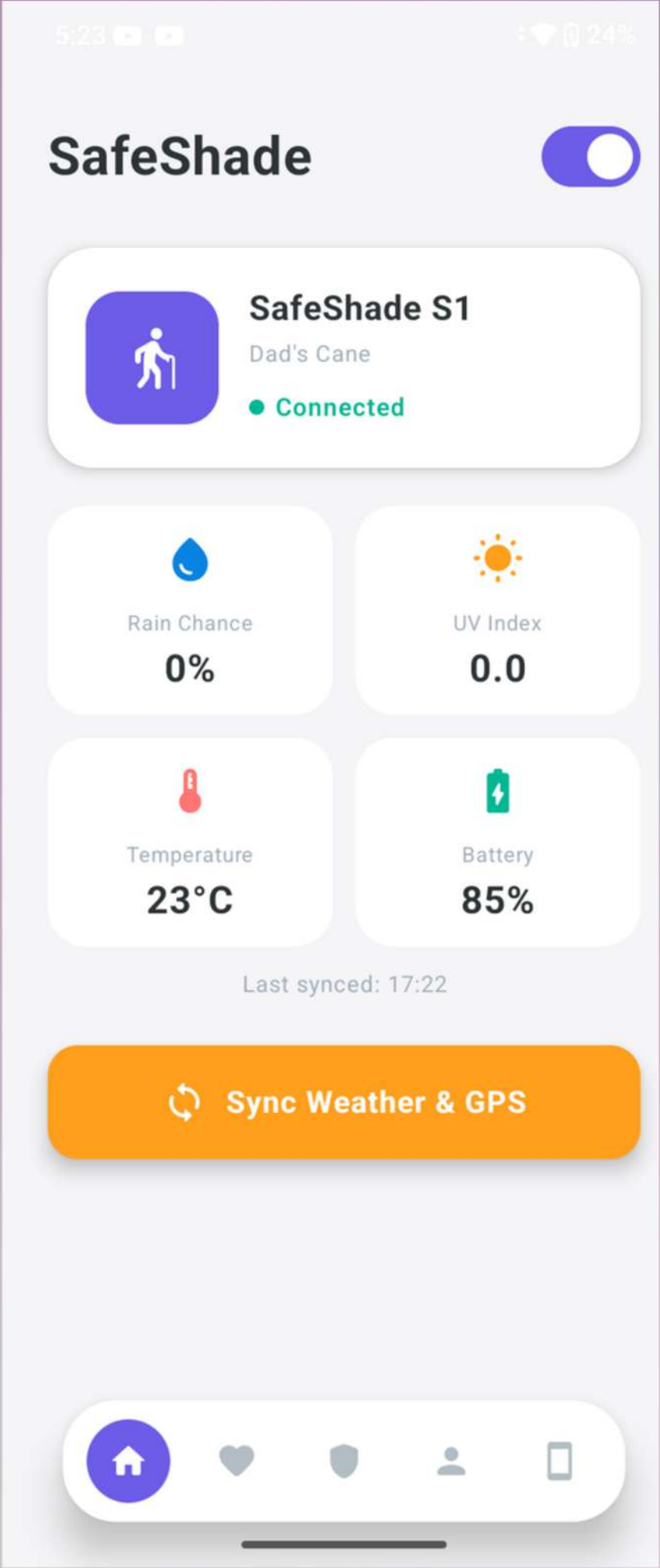


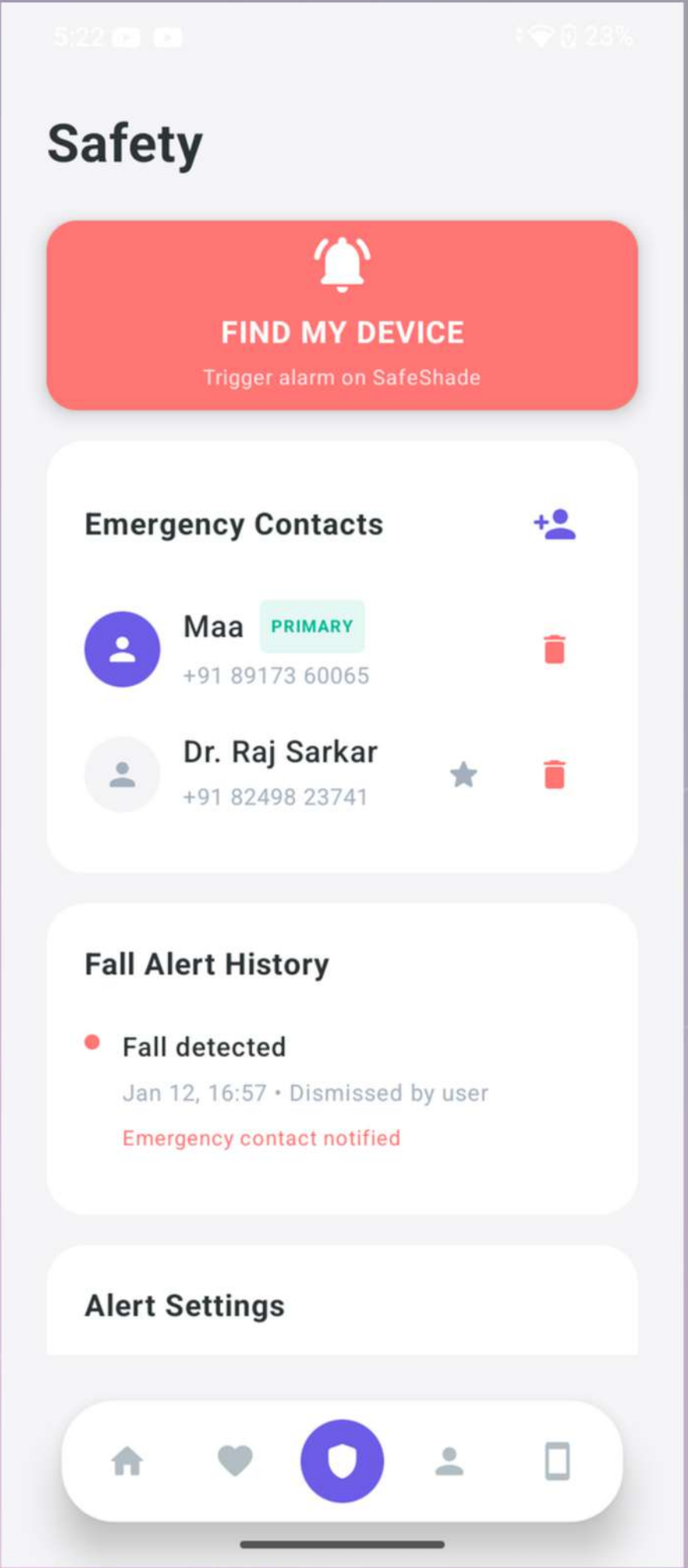
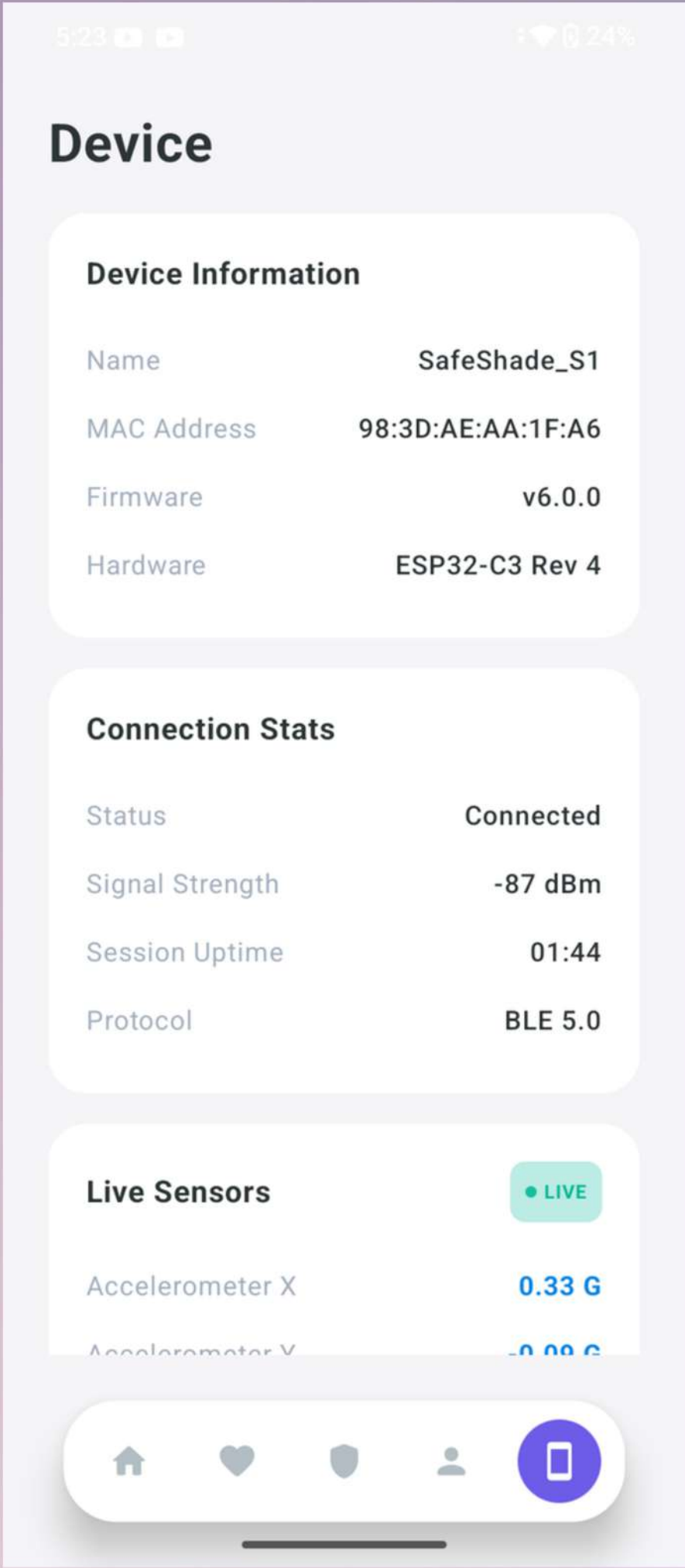
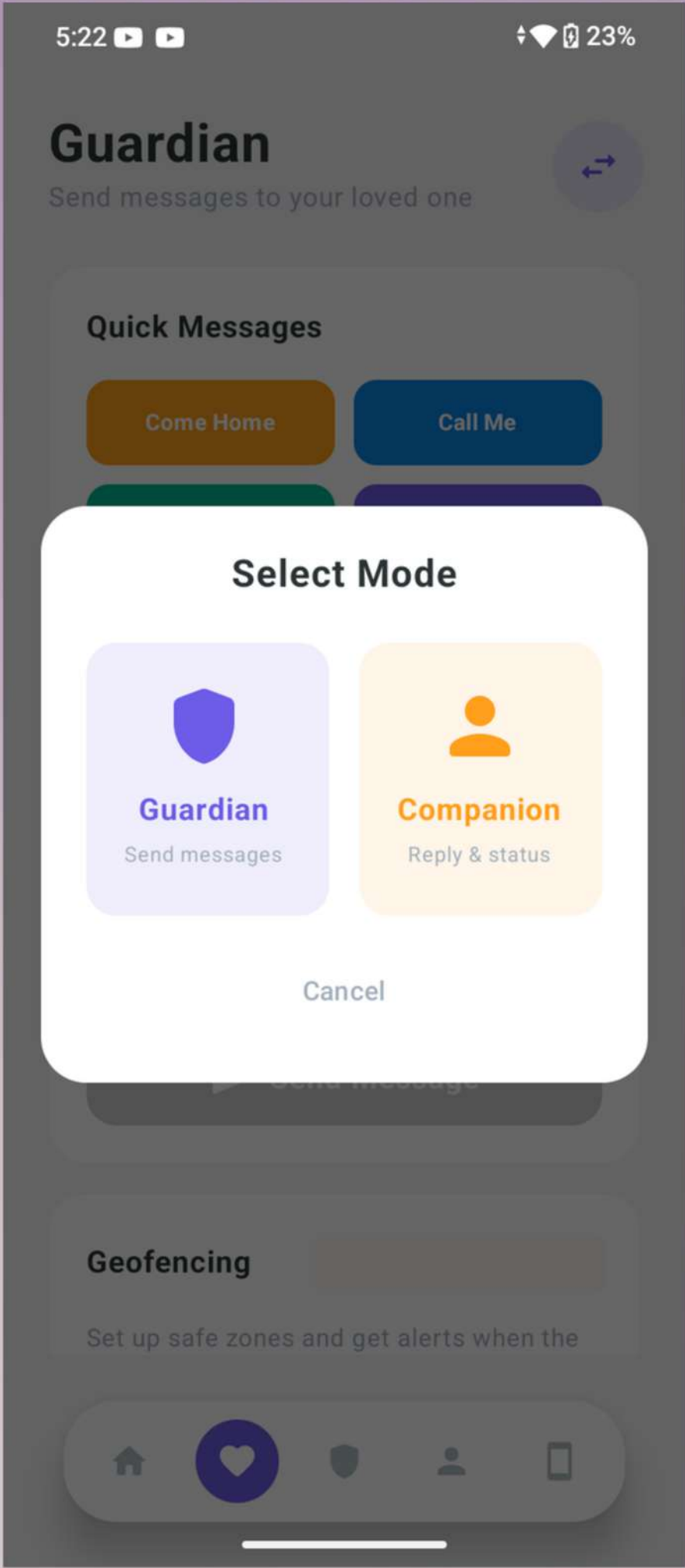
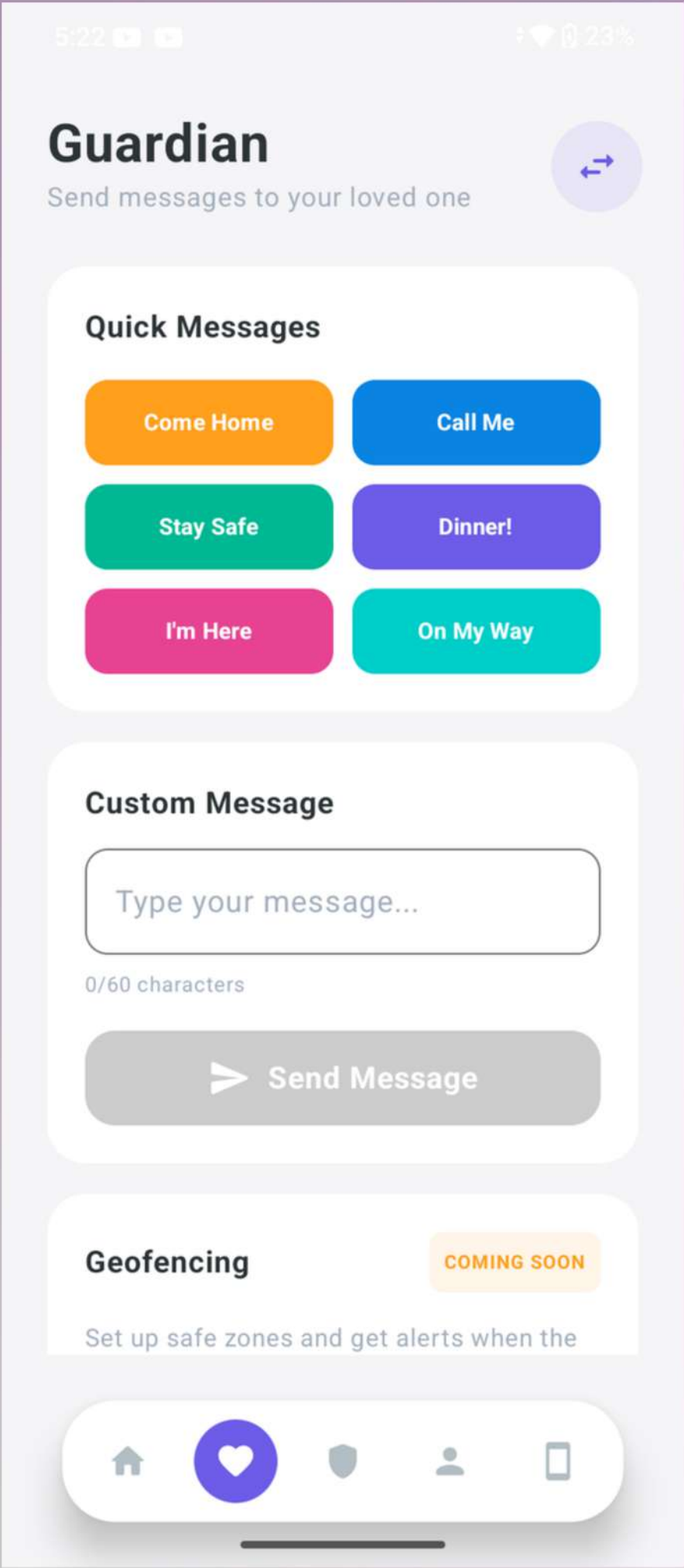
§ Full 5G integration (SIM7670G cellular module), Matter protocol, and health monitoring sensors pending. Planned for PCB v2.0.



v1.3 PCB schematic shown
v2.0 PCB in development.







SafeShade Architecture Diagram

SafeShade S1 Main Device

ESP32-C3 MCU
MPU6050 IMU (fall detection)
NEO-M8N GPS
LDR ambient sensor
WS2812B 270° LEDs
OLED display
SOS button
Rotary mode selector
Active buzzer
Custom PCB v1.3

SafeShade Spark

Nordic nRF9160 SoC
LTE-M / NB-IoT
Independent GPS
NFC Medical ID tag
Panic button
Qi wireless charging
IP68 enclosure

BLE 5.0 for local device communication.
5G/LTE-M for standalone emergency uplink;
independent of phone connectivity.

5G NR Base Station gNB

Mobile Edge Computing (MEC)

AI Fall Detection Model
Real-time gait analysis*
OTA firmware updates
Sub 500ms Emergency Notification*

5G Network Slicing Controller

Emergency QoS slice
Guaranteed bandwidth

5G NR Positioning Engine*

Sub-3m indoor accuracy

C-V2X Module*

PC5 interface
Cyclist-vehicle safety broadcast
300m range

SafeShade Cloud (AWS India)

Guardian dashboard
Safety heat map
Alert history
Community mesh relay
Device management

ERSS 112 India

Direct JSON alert:
name + GPS + medical ID
• fall confidence score

Guardian Android App

BLE alerts
GPS tracking
Medical ID
Messaging
Fall history

Smart City API*

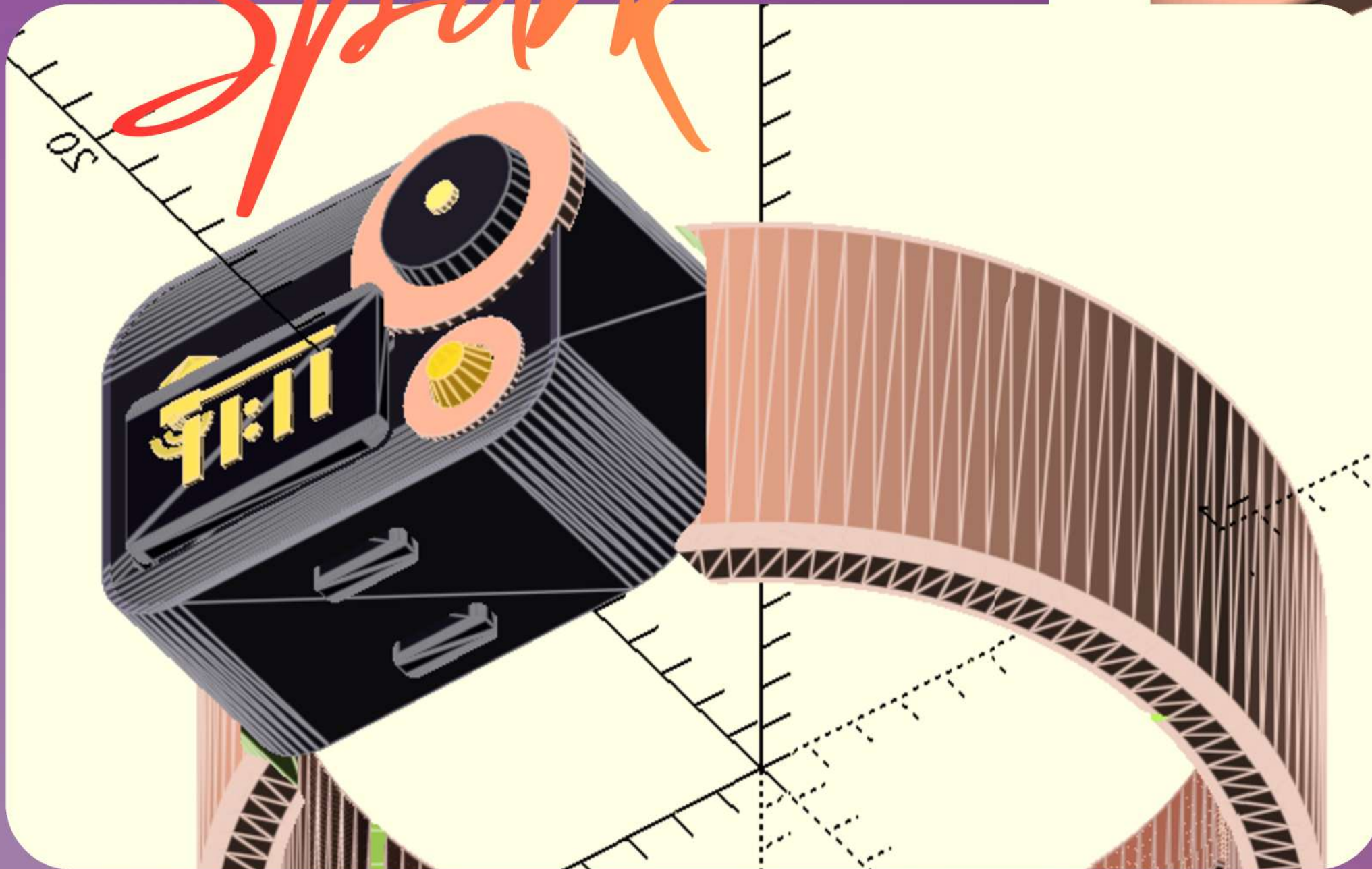
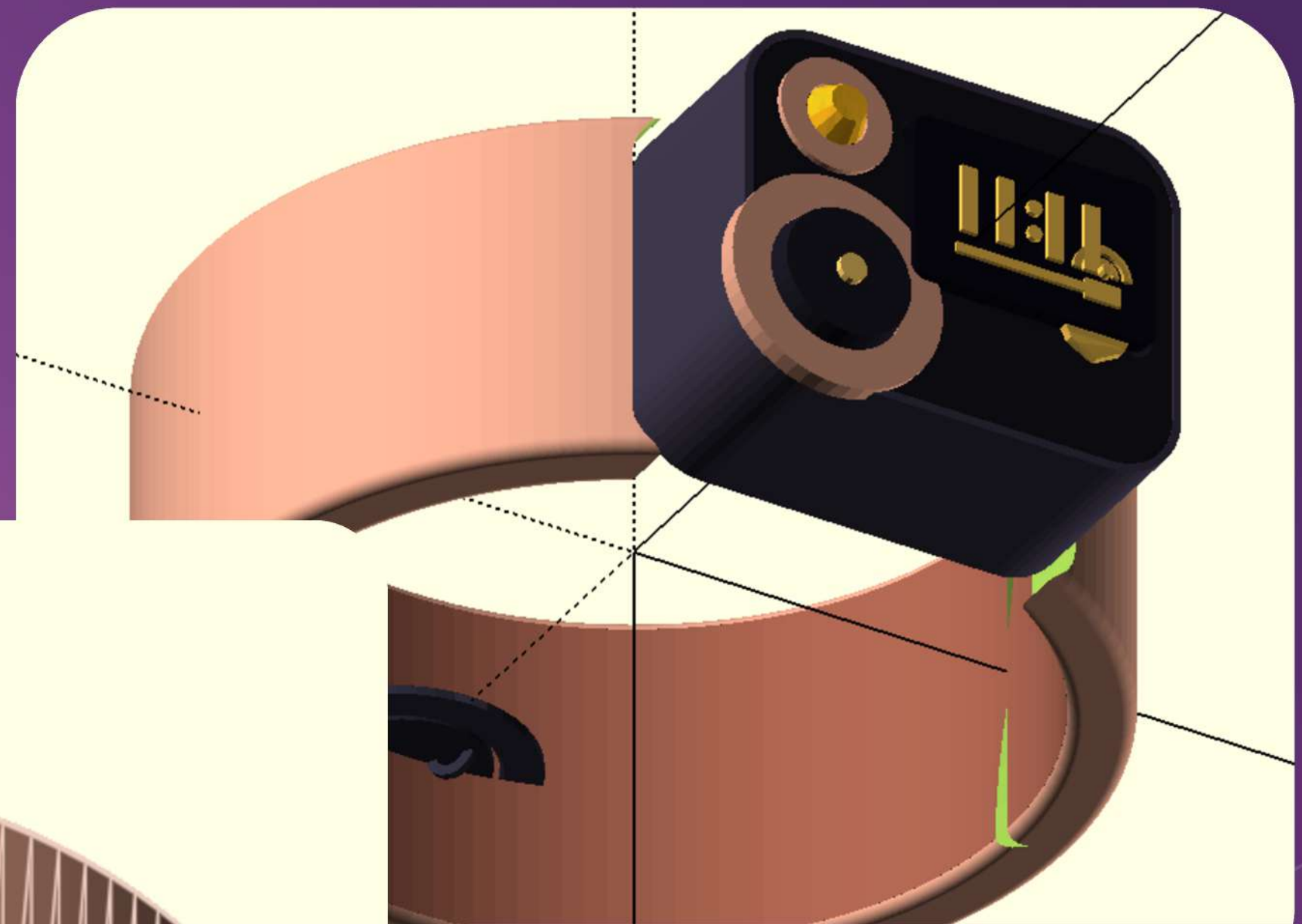
CCTV flag
Street light trigger
City control room

* Planned and/or dependent on infrastructural capability



Safe Shade

Spark



- Independent LTE-M:** calls 112 independently
- Pre-recorded distress message:** plays your name, location, emergency info automatically
- NFC Medical ID:** any smartphone taps the ring and sees your blood type, allergies, contacts. Even if the ring is powered off.
- Anti-removal alert:** detects forceful removal and triggers SOS automatically

The World's First Context-Aware 5G Safety Platform

Patent-pending

Mode	Algorithm Adaptation	UI Changes	Priority Features
 ELDERLY 	Sensitive fall thresholds (2.5g impact)	Large fonts, simplified UI	Fall detection, medication reminders, night bathroom lighting
 KIDS 	Location priority, school-mode scheduling	Colorful theme, parental lock	Geofencing, safe zone alerts, stranger danger SOS
 BIKE 	Vibration filtering, crash detection	Navigation display, speed	Turn-by-turn, brake light, ride tracking
 PET 	Activity patterns (no fall detection)	Owner info always displayed	Virtual leash, activity tracking, lost pet mode
 HELMET	High-speed impact (6g+ threshold)	Minimal distraction UI	Concussion alerts, worker check-ins
 WRIST	Continuous health monitoring	Watch-face style	§ HR, SpO2, sleep tracking, fitness
 BACKPACK 	Balanced all features	Full navigation	Default mode, commuter safety

§ No existing product automatically re-calibrates safety thresholds based on context

Target Demographic

Total Addressable Market

→ **Global Personal Safety: \$25 Billion USD (2025)**

Serviceable Addressable Market

→ **India, UK and SEA: \$5.5 Billion USD**

Segment	Global Size	India Size	Growth
Elderly Care Tech	\$8B	₹6,000 Cr	22% CAGR
Women's Safety	\$3B	₹2,400 Cr	28% CAGR
Kids Safety	\$4B	₹2,000 Cr	18% CAGR
Cycling Safety	\$1.2B	₹450 Cr	18% CAGR
Pet Wearables	\$2.5B	₹680 Cr	14% CAGR

Source | MarketsandMarkets, Elderly Care Market India Report, 2024
NASSCOM, Women Safety Technology Report, 20235.
Grand View Research, Child Safety Market, 2024

**₹1000 crores
opportunity.**

At 1% Indian Market Penetration

Why now?

- ✓ Post-COVID awareness of remote health monitoring
- ✓ 5G/BLE 5.0 enabling reliable connectivity
- ✓ Matter protocol unifying smart home ecosystem
- ✓ India's elderly population growing 3% annually (138M+ people)
- ✓ Women's safety tech gaining government support

SafeShade is better.

Here is the competitive market analysis

Feature	SafeShade	Apple Watch	Life Alert	Garmin Varia	Generic Indian
Price	₹5999 (estimated at scale)	₹29,900 - ₹90,000	₹15,000 setup	₹18,000	₹4000 - ₹9000
Monthly Fee	₹0	₹0	₹2,000/mo	₹0	₹100 - ₹300/mo
5G Implementation	✔ 5G Direct SoS Network Slicing	✔ (Apple Watch Ultra 3 only)	✗	✗	✗
Fall Detection	✔ 3-layer + MEC AI	✔	✔ Basic	✗	✔ Basic
GPS Tracking	✔ Multi-GNSS & 5G Positioning	✔	✔	✗	—
Emergency Calling	✔ Auto voice calls & notify	✔	✔	✗	—
Medical ID	✔ On-device & independent	✔	✗	✗	✗
Multi-User Modes	✔ 7 adaptive modes	✗	✗ Elderly only	✗ Cycling only	✗
Standalone	✔ Standalone + App Support	✔ Standalone + iPhone required	✗	✔	—
LED Visibility	✔ 270°	✗	✗	✔ Rear only	✗
Battery Life	upto 7 days (estimated)	upto 18 hours	30 days	8 hours	7 days

Business Model

1. Hardware (Primary Revenue)

SafeShade S1: ₹5,999

SafeShade Spark: ₹3,999 (planned)

BOM target at 1K units: ~₹2,800

Gross margin target: 45–50%

2. Cloud (Recurring Revenue)

Free tier: All hardware features + 7 day history

SafeShade Plus ₹99/month: Extended history, safety heatmaps, family dashboard, priority ERSS routing

SafeShade Pro ₹299/month: B2B — hospitals, schools, elder care facilities (per-device fleet management)

3. Institutional (B2G/ B2B)

Government tender: Women's safety infrastructure (Nirbhaya Fund eligible)

Elder care facility fleet contracts

School safety programs

Insurance partnerships: Device discount if linked to health insurance policy

 **Grant Use:** 5 MVPs → 8-month pilot → PMF validation → Scale to 20 units

About Us



Dibyendu Ganguly

Dibyendu's DRDO R&D internship showed him how critical, response latency is in emergencies and how far India's safety infrastructure still needs to go. As a third-year Electronics student at KIIT University, he designed SafeShade's hardware, firmware, and cloud architecture from scratch.



Bhoomika Sarkar

Bhoomika's background in hardware mathematics and algorithm modelling is what makes SafeShade's SoS precise & reliable. As a national academic topper and third-year Electronics & CS student at KIIT University, she developed and tuned the context-aware algorithms at the core of the product.

Guided by: Shri. Vishal J. C. (Chief of Staff, I&E, IIT KGP) & Prof. B. N. Rao (School of Electronics, KIIT DU)



Your Everything Safety Companion

684,000 people die from falls every year.
88% of Indian women feel unsafe in public.
45,000 children go missing annually.

None of them have a SafeShade yet.

Thank You & Questions?



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